

Donaldo Buzi

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Experience

Android Engineer

Milan, IT April 2023 – Oct 2024

Hiway Media

- Developed fast and responsive UIs for high-traffic sports streaming apps using Jetpack Compose and Kotlin.
- Contributed to native Android and Android TV apps serving **250,000+ users** (Lazio, SuperTennix, FefTV).
- Engineered performance optimizations resulting in a **20% reduction in crash rates**, **15% less RAM usage**, and a **30% overall performance boost**.
- Implemented a scalable multi-module MVVM architecture, facilitating easier feature expansion and maintenance.

Education

Technical University of Munich (TUM)

Oct 2025 – Present

M.Sc. Computer Science (Erasmus Exchange)

- Focus: Machine Learning, Deep Learning, Systems Programming, Parallel Programming, ML for graphs.

University of Pisa

Sept 2024 – Expected 2026

Master of Science in Computer Science

- **Relevant Coursework:** Formal Languages, Compilers and Interpreters, Principle of Software Composition (Program Analysis), Algorithm Engineering, Algorithm Design, Advanced Programming.

University of Pisa

Sept 2021 – Oct 2024

Bachelor of Science in Computer Science

- **Grade:** 110/110 cum laude (GPA: 4.0/4.0)
- **Thesis:** Comparative Study of ML Indicators for Predicting Cardiac Mortality in Ischemic Heart Disease Patients: github.com/AldoBuzi/buzi_ml4cad 

Projects

MiniImp Compiler

github.com/AldoBuzi/MiniImp 

- Developed a full compiler for a custom imperative language using **OCaml** (Lexer/Parser via Menhir/OCamlLex) to parse code into Abstract Syntax Trees (AST).
- Implemented dataflow analysis (liveness/definition tracking) over Control Flow Graphs (CFG) and register allocation algorithms.
- Generated optimized RISC-V-like assembly code from intermediate representation.

Carta Cultura e Merito (300k+ Users)

 Code |  Play Store

- Developed Italy's most popular utility app for government student bonuses, achieving **300k+ users**.
- Engineered a robust client using Kotlin and Jetpack components (LiveData, ViewModel, Navigation) to handle high-concurrency user sessions during bonus release windows.
- Integrated secure local storage and background services to enhance user retention and utility.

ReddiTrend - Distributed Graph Analytics

github.com/AldoBuzi/ReddiTrend 

- Designed a K8s-native pipeline to extract sentiment and keywords (via **KeyBERT**) from Reddit streams using **Kafka** and **Spark**, persisting structured data to **Cassandra**.
- Orchestrated a production-grade **Kubernetes** cluster using **Strimzi** (Kafka Operator), **Traefik** (Ingress), **Longhorn** (Storage), and **Grafana** for observability.
- Built a high-performance visualization frontend with **React** and **Sigma.js** to render interactive graph nodes, backed by a **FastAPI** microservice.

Technical Skills

Languages: Kotlin, Java, Python, C, OCaml, SQL, Bash.

Frameworks & Tools: Android Jetpack, Kafka, Spark, Docker, Kubernetes, AWS (Lambda/S3), Git, CI/CD.

Concepts: Distributed Systems, Mobile Architecture (MVVM), Compiler Design, Machine Learning.

Languages: Italian (Native), English (Fluent), Spanish (Fluent), Greek.